

Paper Checker – Team Member 1

Module: OCR + Calibration + Confidence Engine

Goal

Build the core AI engine responsible for extracting handwritten text, improving OCR accuracy using calibration, and generating a confidence-aware corrected output.

This module MUST NOT implement Evaluation or FastAPI.

Folder Structure

backend/

services/

ocr/

interface.py

paddle_service.py

ocr_models.py

preprocessing/

image_processor.py

validator.py

config.py

exceptions.py

calibration/

calibration_engine.py

profile.py

correction.py

similarity.py

confidence.py

Tasks

Task 1

Complete OCR Module

Pipeline

Image

↓

Validation

↓

Preprocessing

↓

PaddleOCR

↓

OCRResult

Deliverables

✓ OCRResult

✓ Confidence

✓ Bounding Boxes

Task 2

Calibration Engine

Create a student-specific calibration system.

Input

OCRResult

Calibration Profile

Output

Corrected OCR

Folder

services/calibration/

Task 3

Calibration Profile

Store

Common OCR mistakes

Character Confusions

Word Confusions

Writing Habits

Example

```
{  
  "student_id": "123",  
  "confusions": {  
    "3": "z",
```

```
    "0": "o",

    "5": "s",

    "1": "l"
},

"common_words": [

    "current",

    "resistance",

    "voltage",

    "wizard"
]
}
```

Task 4

Rule-Based OCR Correction

Examples

La3y

↓

Lazy

dos

↓

dog

blaek

↓

black

The correction engine should improve OCR before evaluation.

Task 5

Similarity Engine

Implement

Levenshtein Distance

Fuzzy Matching

Closest Dictionary Match

This will be used to identify the most likely intended word.

Task 6

Confidence Engine

Generate confidence using

OCR Confidence

Calibration Match

Dictionary Match

Correction Confidence

Output

```
{  
  "ocr_confidence":0.79,  
  
  "calibrated_confidence":0.91,  
  
  "overall_confidence":0.88  
}
```

Task 7

GPT OCR Correction

After rule-based correction

Send corrected OCR



GPT



Final Corrected Text

Prompt should

Correct OCR mistakes

Do NOT rewrite answers

Preserve student intent

Task 8

Integration

Final pipeline

Image



Validation



Preprocessing



OCR



Calibration



Confidence



GPT OCR Correction



Corrected OCR

Deliverables

- ✓ OCR Module
 - ✓ Image Validation
 - ✓ Image Preprocessing
 - ✓ Calibration Engine
 - ✓ Student Profile
 - ✓ Rule-Based Corrections
 - ✓ Similarity Engine
 - ✓ Confidence Engine
 - ✓ GPT OCR Correction
-

Branch

feature/calibration-engine